УО «Белорусский государственный университет информатики и радиоэлектроники»
Кафедра ПОИТ

# Отчет по лабораторной работе №5
по предмету
Распределенные вычисления
Вариант 1

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## Задание

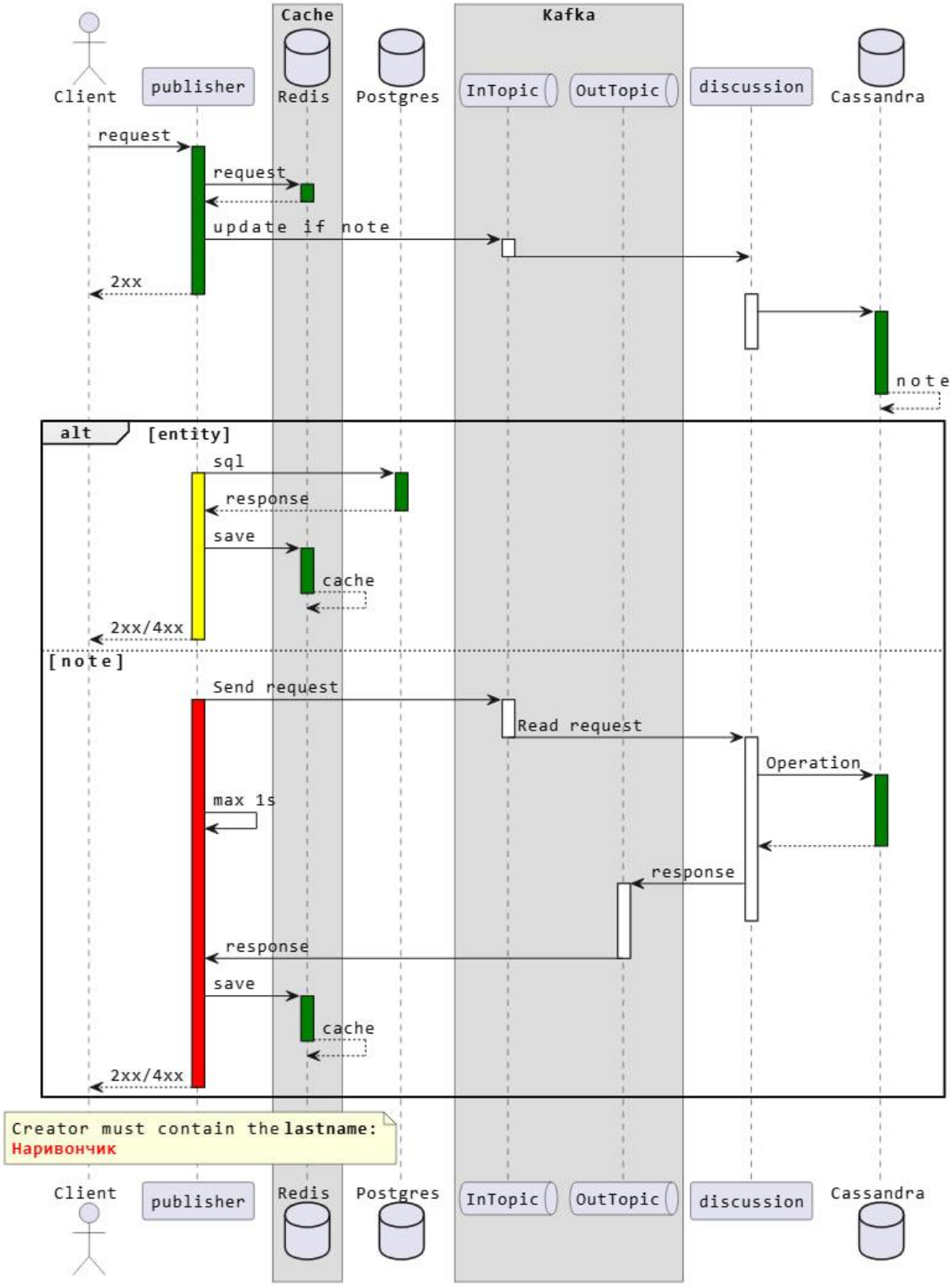
**Дано:**

- Разрабатываемая система обрабатывает сущности **Creator**, **News**, **Mark** и **Note**, которые логически связаны отношениями (см. предыдущие задачи)
  - один-ко-многим (**Creator** и **News**, **News** и **Note**)
  - многие-ко-многим (**News**, **Mark**).
- В **Kafka** настроена передача между модулями **publisher** и **discussion** сущности **Note**.
- **Creator(s)**, **News(s)**, **Mark(s)** хранятся в **Postgres**.
- **Note(s)** хранятся в **Cassandra**.

**Задание**

Необходимо в ранее разработанном распределенном приложении реализовать механизм кеширования в **Redis**.

Основой для выполнения задания является следующая диаграмма взаимодействия




**Технические требования**

- Используйте префикс **/api/v1.0/** для контроллеров REST и их методов,

- Используйте адрес и порт **localhost:24110** для самого приложения (это модуль **publisher**).
- Не отключайте REST **localhost:24130** для модуля **discussion**.
- Используйте обязательный префикс **tbl_** для таблиц(ы) в базе(ах) данных
- Используйте подключение по умолчанию к **Redis** из hub.docker.com

**Рекомендации**

**Изучение технологий**

1. Изучить основы работы с Redis и его возможности.
2. Понять принципы кеширования данных и как Redis может быть использован в качестве кеша.
3. Изучить основные принципы горизонтального масштабирования и как Redis помогает в этом.

**Настройка и установка Redis**

1. Скачать и установить Redis на локальную машину или использовать облачный сервис.
2. Настроить конфигурацию Redis в соответствии с требованиями проекта.
3. Проверить доступность Redis и убедиться, что он работает корректно.

**Интеграция Redis с модулем publisher**

1. Добавить зависимость для поддержки Redis в проекте.
2. Настроить подключение к Redis из модуля **publisher**.
3. Разработать логику кеширования данных в Redis для запросов, требующих повторного использования.
4. Разработать логику обновления данных в Redis и в соответствующих хранилищах для запросов.

**Код**

```
using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

    [ApiController]
    [Route("api/v1.0/[controller]")]
```

```
public class CreatorsController : ControllerBase
{
    private readonly ICreatorService _creatorService;
    private readonly ILogger<CreatorsController> _logger;

    public CreatorsController(ICreatorService creatorService,
ILogger<CreatorsController> logger)
    {
        _creatorService = creatorService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(CreatorResponseTo),
StatusCodes.Status201Created)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<CreatorResponseTo>> CreateCreator(
        [FromBody] CreatorRequestTo createCreatorRequest)
    {
        try
        {
            _logger.LogInformation("Creating creator {request}",
createCreatorRequest);

            CreatorResponseTo createdCreator =
                await _creatorService.CreateCreator(createCreatorRequest);

            return CreatedAtAction(
                nameof(CreateCreator),
                new { id = createdCreator.Id },
                createdCreator
            );
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error creating creator");
            return StatusCode(500, "An error occurred while creating the creator");
        }
    }
```

```
        [HttpGet]
        [ProducesResponseType(typeof(IEnumerable<CreatorResponseTo>),
StatusCodes.Status200OK)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<IEnumerable<CreatorResponseTo>>>
GetAllCreators()
        {
            try
            {
                _logger.LogInformation("Getting all creators");

                IEnumerable<CreatorResponseTo> creators =
                    await _creatorService.GetAllCreators();

                return Ok(creators);
            }
            catch (Exception ex)
            {
                _logger.LogError(ex, "Error getting all creators");
                return StatusCode(500, "An error occurred while retrieving creators");
            }
        }

        [HttpGet("{id:long}")]
        [ProducesResponseType(typeof(CreatorResponseTo),
StatusCodes.Status200OK)]
        [ProducesResponseType(StatusCodes.Status404NotFound)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<CreatorResponseTo>>
GetCreatorById([FromRoute] long id)
        {
            try
            {
                _logger.LogInformation("Getting creator by id: {Id}", id);

                var getCreatorRequest = new CreatorRequestTo { Id = id };
                CreatorResponseTo creator =
                    await _creatorService.GetCreator(getCreatorRequest);
```

```
            return Ok(creator);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "Creator not found with id: {Id}", id);
            return NotFound($"Creator with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting creator by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the creator");
        }
    }

    [HttpPut]
    [ProducesResponseType(typeof(CreatorResponseTo),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<CreatorResponseTo>> UpdateCreator(
        [FromBody] CreatorRequestTo updateCreatorRequest)
    {
        try
        {
            _logger.LogInformation(
                "Updating creator with id: {Id}",
                updateCreatorRequest.Id);

            var updatedCreator =
                await _creatorService.UpdateCreator(updateCreatorRequest);

            return Ok(updatedCreator);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(
                ex,
                "Creator not found for update with id: {Id}",
                updateCreatorRequest.Id);
```

```
            return NotFound(
                $"Creator with id {updateCreatorRequest.Id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(
                ex,
                "Error updating creator with id: {Id}",
                updateCreatorRequest.Id);

            return StatusCode(
                500,
                "An error occurred while updating the creator");
        }
    }

    [HttpDelete("{id:long}")]
    [ProducesResponseType(StatusCodes.Status204NoContent)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<IActionResult> DeleteCreator([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Deleting creator with id: {Id}", id);

            var deleteCreatorRequest = new CreatorRequestTo { Id = id };
            await _creatorService.DeleteCreator(deleteCreatorRequest);

            return NoContent();
        }
        catch (CreatorNotFoundException ex)
        {
            _logger.LogWarning(
                ex,
                "Creator not found for deletion with id: {Id}",
                id);

            return NotFound();
```

```
            }
            catch (Exception ex)
            {
                _logger.LogError(
                    ex,
                    "Error deleting creator with id: {Id}",
                    id);

                return StatusCode(
                    500,
                    "An error occurred while deleting the creator");
            }
        }
    }

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

[ApiController]
[Route("api/v1.0/[controller]")]
public class MarksController : ControllerBase
{
    private readonly IMarkService _markService;
    private readonly ILogger<MarksController> _logger;

    public MarksController(IMarkService markService, ILogger<MarksController>
logger)
    {
        _markService = markService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(MarkResponseTo),
StatusCodes.Status201Created)]
```

```
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<MarkResponseTo>> CreateMark(
        [FromBody] MarkRequestTo createMarkRequest)
    {
        try
        {
            _logger.LogInformation("Creating mark {request}", createMarkRequest);

            MarkResponseTo createdMark =
                await _markService.CreateMark(createMarkRequest);

            return CreatedAtAction(
                nameof(CreateMark),
                new { id = createdMark.Id },
                createdMark
            );
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error creating mark");
            return StatusCode(500, "An error occurred while creating the mark");
        }
    }

    [HttpGet]
    [ProducesResponseType(typeof(IEnumerable<MarkResponseTo>),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<IEnumerable<MarkResponseTo>>>
GetAllMarks()
    {
        try
        {
            _logger.LogInformation("Getting all marks");

            IEnumerable<MarkResponseTo> marks =
                await _markService.GetAllMarks();

            return Ok(marks);
```

```
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting all marks");
            return StatusCode(500, "An error occurred while retrieving marks");
        }
    }

    [HttpGet("{id:long}")]
    [ProducesResponseType(typeof(MarkResponseTo), StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<MarkResponseTo>>
GetMarkById([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Getting mark by id: {Id}", id);

            var getMarkRequest = new MarkRequestTo { Id = id };
            MarkResponseTo mark =
                await _markService.GetMark(getMarkRequest);

            return Ok(mark);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "Mark not found with id: {Id}", id);
            return NotFound($"Mark with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting mark by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the mark");
        }
    }

    [HttpPut]
    [ProducesResponseType(typeof(MarkResponseTo), StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
```

```
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<MarkResponseTo>> UpdateMark(
        [FromBody] MarkRequestTo updateMarkRequest)
    {
        try
        {
            _logger.LogInformation("Updating mark with id: {Id}",
updateMarkRequest.Id);

            var updatedMark =
                await _markService.UpdateMark(updateMarkRequest);

            return Ok(updatedMark);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(
                ex,
                "Mark not found for update with id: {Id}",
                updateMarkRequest.Id);

            return NotFound(
                $"Mark with id {updateMarkRequest.Id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(
                ex,
                "Error updating mark with id: {Id}",
                updateMarkRequest.Id);

            return StatusCode(
                500,
                "An error occurred while updating the mark");
        }
    }

    [HttpDelete("{id:long}")]
    [ProducesResponseType(StatusCodes.Status204NoContent)]
```

```
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<IActionResult> DeleteMark([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Deleting mark with id: {Id}", id);

            var deleteMarkRequest = new MarkRequestTo { Id = id };
            await _markService.DeleteMark(deleteMarkRequest);

            return NoContent();
        }
        catch (MarkNotFoundException ex)
        {
            _logger.LogWarning(ex, "Mark not found for deletion with id: {Id}", id);
            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error deleting mark with id: {Id}", id);
            return StatusCode(500, "An error occurred while deleting the mark");
        }
    }
}

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

[ApiController]
[Route("api/v1.0/[controller]")]
public class NewsController : ControllerBase
{
    private readonly INewsService _newsService;
    private readonly ILogger<NewsController> _logger;
```

```
    public NewsController(INewsService newsService, ILogger<NewsController>
logger)
    {
        _newsService = newsService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(NewsResponseTo),
StatusCodes.Status201Created)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NewsResponseTo>>
CreateNews([FromBody] NewsRequestTo createNewsRequest)
    {
        try
        {
            _logger.LogInformation("Creating news with title: {Title}",
createNewsRequest.Title);

            var createdNews = await _newsService.CreateNews(createNewsRequest);

            return CreatedAtAction(
                nameof(GetNewsById),
                new { id = createdNews.Id },
                createdNews
            );
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error creating news");
            return StatusCode(500, "An error occurred while creating the news");
        }
    }

    [HttpGet]
    [ProducesResponseType(typeof(IEnumerable<NewsResponseTo>),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
```

```
    public async Task<ActionResult<IEnumerable<NewsResponseTo>>>
GetAllNews()
    {
        try
        {
            _logger.LogInformation("Getting all news");

            var news = await _newsService.GetAllNews();

            return Ok(news);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting all news");
            return StatusCode(500, "An error occurred while retrieving news");
        }
    }

    [HttpGet("{id:long}")]
    [ProducesResponseType(typeof(NewsResponseTo), StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NewsResponseTo>>
GetNewsById([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Getting news by id: {Id}", id);

            var getNewsRequest = new NewsRequestTo { Id = id };
            var news = await _newsService.GetNews(getNewsRequest);

            return Ok(news);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "News not found with id: {Id}", id);
            return NotFound($"News with id {id} not found");
        }
        catch (Exception ex)
```

```
        {
            _logger.LogError(ex, "Error getting news by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the news");
        }
    }

    [HttpPut]
    [ProducesResponseType(typeof(NewsResponseTo), StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NewsResponseTo>> UpdateNews(
        [FromBody] NewsRequestTo updateNewsRequest)
    {
        try
        {
            _logger.LogInformation("Updating news with id: {Id}",
updateNewsRequest.Id);

            var updatedNews = await
_newsService.UpdateNews(updateNewsRequest);

            return Ok(updatedNews);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "News not found for update with id: {Id}",
updateNewsRequest.Id);
            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error updating news with id: {Id}",
updateNewsRequest.Id);
            return StatusCode(500, "An error occurred while updating the news");
        }
    }

    [HttpDelete("{id:long}")]
    [ProducesResponseType(StatusCodes.Status204NoContent)]
```

```
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<IActionResult> DeleteNews([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Deleting news with id: {Id}", id);

            var deleteNewsRequest = new NewsRequestTo { Id = id };
            await _newsService.DeleteNews(deleteNewsRequest);

            return NoContent();
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "News not found for deletion with id: {Id}", id);
            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error deleting news with id: {Id}", id);
            return StatusCode(500, "An error occurred while deleting the news");
        }
    }
}

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

    [ApiController]
    [Route("api/v1.0/[controller]")]
    public class NotesController : ControllerBase
    {
        private readonly INoteService _noteService;
        private readonly ILogger<NotesController> _logger;
```

```
    public NotesController(INoteService noteService, ILogger<NotesController>
logger)
    {
        _noteService = noteService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status201Created)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NoteResponseTo>> CreateNote(
        [FromBody] NoteRequestTo createNoteRequest)
    {
        try
        {
            _logger.LogInformation("Creating note {request}", createNoteRequest);

            NoteResponseTo createdNote =
                await _noteService.CreateNote(createNoteRequest);

            return CreatedAtAction(
                nameof(CreateNote),
                new { id = createdNote.Id },
                createdNote
            );
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error creating note");
            return StatusCode(500, "An error occurred while creating the note");
        }
    }

    [HttpGet]
    [ProducesResponseType(typeof(IEnumerable<NoteResponseTo>),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
```

```
    public async Task<ActionResult<IEnumerable<NoteResponseTo>>>
GetAllNotes()
    {
        try
        {
            _logger.LogInformation("Getting all notes");

            IEnumerable<NoteResponseTo> notes =
                await _noteService.GetAllNotes();

            return Ok(notes);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting all notes");
            return StatusCode(500, "An error occurred while retrieving notes");
        }
    }

    [HttpGet("{id:long}")]
    [ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NoteResponseTo>>
GetNoteById([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Getting note by id: {Id}", id);

            var getNoteRequest = new NoteRequestTo { Id = id };
            NoteResponseTo note =
                await _noteService.GetNote(getNoteRequest);

            return Ok(note);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "Note not found with id: {Id}", id);
```

```
            return NotFound($"Note with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting note by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the note");
        }
    }

    [HttpPut]
    [ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NoteResponseTo>> UpdateNote(
        [FromBody] NoteRequestTo updateNoteRequest)
    {
        try
        {
            _logger.LogInformation("Updating note with id: {Id}",
updateNoteRequest.Id);

            var updatedNote =
                await _noteService.UpdateNote(updateNoteRequest);

            return Ok(updatedNote);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(
                ex,
                "Note not found for update with id: {Id}",
                updateNoteRequest.Id);

            return NotFound(
                $"Note with id {updateNoteRequest.Id} not found");
        }
        catch (Exception ex)
        {
```

```
            _logger.LogError(
                ex,
                "Error updating note with id: {Id}",
                updateNoteRequest.Id);

            return StatusCode(
                500,
                "An error occurred while updating the note");
        }
    }

    [HttpDelete("{id:long}")]
    [ProducesResponseType(StatusCodes.Status204NoContent)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<IActionResult> DeleteNote([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Deleting note with id: {Id}", id);

            var deleteNoteRequest = new NoteRequestTo { Id = id };
            await _noteService.DeleteNote(deleteNoteRequest);

            return NoContent();
        }
        catch (NoteNotFoundException ex)
        {
            _logger.LogWarning(ex, "Note not found for deletion with id: {Id}", id);
            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error deleting note with id: {Id}", id);
            return StatusCode(500, "An error occurred while deleting the note");
        }
    }
}

using System.ComponentModel.DataAnnotations;
```

```
namespace rest1.application.DTOs.requests;

public class CreatorRequestTo
{
	public long? Id { get; set; }

	[StringLength(64, MinimumLength = 2)]
	public string? Login { get; set; }

	[StringLength(128, MinimumLength = 8)]
	public string? Password { get; set; }

	[StringLength(64, MinimumLength = 2)]
	public string? Firstname { get; set; }

	[StringLength(64, MinimumLength = 2)]
	public string? Lastname { get; set; }
}
```

```
using System.ComponentModel.DataAnnotations;

namespace rest1.application.DTOs.requests;

public class MarkRequestTo
{
   public long? Id { get; set; }

   [StringLength(32, MinimumLength = 2)]
   public string? Name { get; set; }

   public long[] NewsIds { get; set; }
}
```

```
using System.ComponentModel.DataAnnotations;

namespace rest1.application.DTOs.requests;

public class NewsRequestTo
{
```

```
    public long? Id { get; set; }

    public long CreatorId { get; set; }

    [StringLength(64, MinimumLength = 2)]
    public string Title { get; set; } = string.Empty;

    [StringLength(2048, MinimumLength = 4)]
    public string Content { get; set; } = string.Empty;

    public long[] MarkIds { get; set; }
}

using System.ComponentModel.DataAnnotations;

namespace rest1.application.DTOs.requests;

public class NoteRequestTo
{
    public long? Id { get; set; }

    public long NewsId { get; set; }

    [StringLength(2048, MinimumLength = 2)]
    public string Content { get; set; }
}

namespace rest1.application.DTOs.responses;

public class CreatorResponseTo (
    string login,
    string password,
    string firstName,
    string lastName)
{
    public long Id { get; set; }

    public string Login { get; set; } = login;

    public string Password { get; set; } = password;
```

```
    public string Firstname { get; set; } = firstName;

    public string Lastname { get; set; } = lastName;

}

namespace rest1.application.DTOs.responses;

public class MarkResponseTo(
        long id,
        string name)
{
    public long? Id { get; set; } = id;

    public string? Name { get; set; } = name;
}

namespace rest1.application.DTOs.responses;

public class NewsResponseTo(
    string title,
    string content,
    DateTime createdAt,
    DateTime modified)
{
    public long Id { get; set; }

    public long CreatorId { get; set; }

    public string Title { get; set; } = title;

    public string Content { get; set; } = content;

    public DateTime CreatedAt { get; set; } = createdAt;

    public DateTime Modified { get; set; } = modified;
}

namespace rest1.application.DTOs.responses;
```

```
public class NoteResponseTo(
        long id,
        long newsId,
        string content)
{
    public long Id { get; set; } = id;

    public long NewsId { get; set; } = newsId;

    public string Content {  get; set; } = content;
}

namespace rest1.application.exceptions;

public class CreatorNotFoundException : NotFoundException
{
    public CreatorNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public CreatorNotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class MarkNotFoundException : NotFoundException
{
    public MarkNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public MarkNotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class NewNotFoundException : NotFoundException
{
    public NewNotFoundException(string message, Exception inner) :
base(message, inner) { }
```

```
    public NewNotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class NotFoundException : Exception
{
    public NotFoundException(string message, Exception inner) : base(message,
inner) { }

    public NotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class NoteNotFoundException : NotFoundException
{
    public NoteNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public NoteNotFoundException(string message) : base(message) { }
}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface ICreatorRepository : IRepository<Creator>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface IMarkRepository : IRepository<Mark>
{

}
```

```
using rest1.core.entities;

namespace rest1.application.interfaces;

public interface INewsRepository : IRepository<News>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface INoteRepository : IRepository<Note>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface IRepository<TEntity> where TEntity : Entity
{
    //get records
    Task<TEntity?> GetByIdAsync(long id, CancellationToken cancellationToken
= default);
    Task<IEnumerable<TEntity>> GetAllAsync(CancellationToken ct = default);

    //create record
    Task<TEntity> AddAsync(TEntity entity, CancellationToken ct = default);

    //update record
    Task<TEntity?> UpdateAsync(TEntity entity, CancellationToken ct = default);

    //delete record
    Task<TEntity?> DeleteAsync(TEntity entity, CancellationToken ct = default);

    //find records
```

```
    Task<TEntity?> FindAsync(ISpecification<TEntity> spec, CancellationToken
ct = default);
    Task<IEnumerable<TEntity>> FindAllAsync(ISpecification<TEntity> spec,
CancellationToken ct = default);

    //count records
    Task<int> CountAsync(ISpecification<TEntity> spec, CancellationToken ct =
default);

    //is any record
    Task<bool> AnyAsync(ISpecification<TEntity> spec, CancellationToken ct =
default);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface ICreatorService
{
    Task<CreatorResponseTo> CreateCreator(CreatorRequestTo
createCreatorRequestTo);

    Task<IEnumerable<CreatorResponseTo>> GetAllCreators();

    Task<CreatorResponseTo> GetCreator(CreatorRequestTo
getCreatorRequestTo);

    Task<CreatorResponseTo> UpdateCreator(CreatorRequestTo
updateCreatorRequestTo);

    Task DeleteCreator(CreatorRequestTo deleteCreatorRequestTo);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;
```

```
public interface IMarkService
{
    Task<MarkResponseTo> CreateMark(MarkRequestTo createMarkRequestTo);

    Task<IEnumerable<MarkResponseTo>> GetAllMarks();

    Task<MarkResponseTo> GetMark(MarkRequestTo getMarkRequestTo);

    Task<MarkResponseTo> UpdateMark(MarkRequestTo updateMarkRequestTo);

    Task DeleteMark(MarkRequestTo deleteMarkRequestTo);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface INewsService
{
    Task<NewsResponseTo> CreateNews(NewsRequestTo createNewsRequestTo);

    Task<IEnumerable<NewsResponseTo>> GetAllNews();

    Task<NewsResponseTo> GetNews(NewsRequestTo getNewsRequestTo);

    Task<NewsResponseTo> UpdateNews(NewsRequestTo
updateNewsRequestTo);

    Task DeleteNews(NewsRequestTo deleteNewsRequestTo);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface INoteService
{
    Task<NoteResponseTo> CreateNote(NoteRequestTo createNoteRequestTo);
```

```
    Task<IEnumerable<NoteResponseTo>> GetAllNotes();

    Task<NoteResponseTo> GetNote(NoteRequestTo getNoteRequestTo);

    Task<NoteResponseTo> UpdateNote(NoteRequestTo updateNoteRequestTo);

    Task DeleteNote(NoteRequestTo deleteNoteRequestTo);
}

using System.Linq.Expressions;

namespace rest1.application.interfaces;

public interface ISpecification<T>
{
    //criteria expression
    Expression<Func<T, bool>> Criteria { get; }

    //join expressions
    List<Expression<Func<T, object>>> Includes { get; }

    //order by expressions
    Expression<Func<T, object>>? OrderBy { get; }
    Expression<Func<T, object>>? OrderByDescending { get; }

    //taken records count
    int? Take { get; }

    //skipped records count
    int? Skip { get; }

    //pagination flag
    bool IsPagingEnabled { get; }
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;
using AutoMapper;
```

```
namespace rest1.application.mappers;

public class CreatorProfile : Profile
{
   public CreatorProfile()
   {
      CreateMap<CreatorRequestTo, Creator>()
         .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
         .ForMember(dest => dest.Login, opt => opt.MapFrom(src => src.Login))
         .ForMember(dest => dest.Firstname, opt => opt.MapFrom(src =>
src.Firstname))
         .ForMember(dest => dest.Lastname, opt => opt.MapFrom(src =>
src.Lastname))
         .ForMember(dest => dest.Password, opt => opt.MapFrom(src =>
src.Password));

      CreateMap<Creator, CreatorResponseTo>()
         .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
         .ForMember(dest => dest.Login, opt => opt.MapFrom(src => src.Login))
         .ForMember(dest => dest.Firstname, opt => opt.MapFrom(src =>
src.Firstname))
         .ForMember(dest => dest.Lastname, opt => opt.MapFrom(src =>
src.Lastname))
         .ForMember(dest => dest.Password, opt => opt.MapFrom(src =>
src.Password));
   }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;

namespace rest1.application.mappers;

public class MarkProfile : Profile
{
   public MarkProfile()
   {
```

```
        CreateMap<MarkRequestTo, Mark>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.Name, opt => opt.MapFrom(src => src.Name));

        CreateMap<Mark, MarkResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.Name, opt => opt.MapFrom(src => src.Name));
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;

namespace rest1.application.mappers;

public class NewsProfile : Profile
{
    public NewsProfile()
    {
        CreateMap<NewsRequestTo, News>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.CreatedAt, opt => opt.MapFrom(src =>
DateTime.UtcNow))
            .ForMember(dest => dest.Modified, opt => opt.MapFrom(src =>
DateTime.UtcNow))
            .ForMember(dest => dest.CreatorId, opt => opt.MapFrom(src =>
src.CreatorId))
            .ForMember(dest => dest.Content, opt => opt.MapFrom(src =>
src.Content))
            .ForMember(dest => dest.Title, opt => opt.MapFrom(src => src.Title));

        CreateMap<News, NewsResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.CreatorId, opt => opt.MapFrom(src =>
src.CreatorId))
            .ForMember(dest => dest.Title, opt => opt.MapFrom(src => src.Title))
            .ForMember(dest => dest.Content, opt => opt.MapFrom(src =>
src.Content))
```

```
            .ForMember(dest => dest.CreatedAt, opt => opt.MapFrom(src =>
src.CreatedAt))
            .ForMember(dest => dest.Modified, opt => opt.MapFrom(src =>
src.Modified));
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;

namespace rest1.application.mappers;

public class NoteProfile : Profile
{
    public NoteProfile()
    {
        CreateMap<NoteRequestTo, Note>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.NewsId, opt => opt.MapFrom(src =>
src.NewsId));

        CreateMap<Note, NoteResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.NewsId, opt => opt.MapFrom(src =>
src.NewsId));
    }
}

using AutoMapper;
using Microsoft.AspNetCore.Components.Forms;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;
```

```
public class CreatorService : ICreatorService
{
    private readonly IMapper _mapper;
    private readonly ICreatorRepository _creatorRepository;

    public CreatorService(IMapper mapper, ICreatorRepository repository)
    {
        _mapper = mapper;
        _creatorRepository = repository;
    }

    public async Task<CreatorResponseTo> CreateCreator(CreatorRequestTo
createCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(createCreatorRequestTo);

        Creator createdCreator = await
_creatorRepository.AddAsync(creatorFromDto);

        CreatorResponseTo dtoFromCreatedEditor =
_mapper.Map<CreatorResponseTo>(createdCreator);

        return dtoFromCreatedEditor;
    }

    public async Task DeleteCreator(CreatorRequestTo deleteCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(deleteCreatorRequestTo);

        _ = await _creatorRepository.DeleteAsync(creatorFromDto)
            ?? throw new CreatorNotFoundException(
                $"Delete creator {creatorFromDto} was not found");
    }

    public async Task<IEnumerable<CreatorResponseTo>> GetAllCreators()
    {
        IEnumerable<Creator> allCreators =
            await _creatorRepository.GetAllAsync();
```

```
        var response = new List<CreatorResponseTo>();

        foreach (Creator creator in allCreators)
        {
            response.Add(_mapper.Map<CreatorResponseTo>(creator));
        }

        return response;
    }

    public async Task<CreatorResponseTo> GetCreator(CreatorRequestTo
getCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(getCreatorRequestTo);

        Creator demandedCreator =
            await _creatorRepository.GetByIdAsync(creatorFromDto.Id)
            ?? throw new CreatorNotFoundException(
                $"Demanded creator {creatorFromDto} was not found");

        return _mapper.Map<CreatorResponseTo>(demandedCreator);
    }

    public async Task<CreatorResponseTo> UpdateCreator(CreatorRequestTo
updateCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(updateCreatorRequestTo);

        Creator updatedCreator =
            await _creatorRepository.UpdateAsync(creatorFromDto)
            ?? throw new CreatorNotFoundException(
                $"Update creator {creatorFromDto} was not found");

        return _mapper.Map<CreatorResponseTo>(updatedCreator);
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
```

```
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;

public class MarkService : IMarkService
{
    private readonly IMapper _mapper;
    private readonly IMarkRepository _markRepository;

    public MarkService(IMapper mapper, IMarkRepository repository)
    {
        _mapper = mapper;
        _markRepository = repository;
    }

    public async Task<MarkResponseTo> CreateMark(MarkRequestTo
createMarkRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(createMarkRequestTo);

        Mark createdMark = await _markRepository.AddAsync(markFromDto);

        MarkResponseTo dtoFromCreatedMark =
            _mapper.Map<MarkResponseTo>(createdMark);

        return dtoFromCreatedMark;
    }

    public async Task DeleteMark(MarkRequestTo deleteMarkRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(deleteMarkRequestTo);

        _ = await _markRepository.DeleteAsync(markFromDto)
            ?? throw new MarkNotFoundException(
                $"Delete mark {markFromDto} was not found");
    }
```

```
    public async Task<IEnumerable<MarkResponseTo>> GetAllMarks()
    {
        IEnumerable<Mark> allMarks =
            await _markRepository.GetAllAsync();

        var allMarksResponseTos = new List<MarkResponseTo>();

        foreach (Mark mark in allMarks)
        {
            MarkResponseTo markTo =
                _mapper.Map<MarkResponseTo>(mark);

            allMarksResponseTos.Add(markTo);
        }

        return allMarksResponseTos;
    }

    public async Task<MarkResponseTo> GetMark(MarkRequestTo
getMarksRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(getMarksRequestTo);

        Mark demandedMark =
            await _markRepository.GetByIdAsync(markFromDto.Id)
            ?? throw new MarkNotFoundException(
                $"Demanded mark {markFromDto} was not found");

        MarkResponseTo demandedMarkResponseTo =
            _mapper.Map<MarkResponseTo>(demandedMark);

        return demandedMarkResponseTo;
    }

    public async Task<MarkResponseTo> UpdateMark(MarkRequestTo
updateMarkRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(updateMarkRequestTo);

        Mark updateMark =
```

```
            await _markRepository.UpdateAsync(markFromDto)
            ?? throw new MarkNotFoundException(
                $"Update mark {markFromDto} was not found");

        MarkResponseTo updateMarkResponseTo =
            _mapper.Map<MarkResponseTo>(updateMark);

        return updateMarkResponseTo;
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;

 public class NewsService : INewsService
    {
        private readonly IMapper _mapper;

        private readonly INewsRepository _newsRepository;

        public NewsService(IMapper mapper, INewsRepository repository)
        {
            _mapper = mapper;
            _newsRepository = repository;
        }

        public async Task<NewsResponseTo> CreateNews(NewsRequestTo
createNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(createNewsRequestTo);

            News createdNews = await _newsRepository.AddAsync(newsFromDto);
```

```
            NewsResponseTo dtoFromCreatedNews =
_mapper.Map<NewsResponseTo>(createdNews);

            return dtoFromCreatedNews;
        }

        public async Task DeleteNews(NewsRequestTo deleteNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(deleteNewsRequestTo);
            _ = await _newsRepository.DeleteAsync(newsFromDto) ?? throw new
NewNotFoundException($"Deleted news {newsFromDto} was not found");
        }

        public async Task<IEnumerable<NewsResponseTo>> GetAllNews()
        {
            IEnumerable<News> allNews = await _newsRepository.GetAllAsync();

            var allNewsResponseTos = new List<NewsResponseTo>();
            foreach (News news in allNews)
            {
                NewsResponseTo newsTo = _mapper.Map<NewsResponseTo>(news);
                allNewsResponseTos.Add(newsTo);
            }

            return allNewsResponseTos;
        }

        public async Task<NewsResponseTo> GetNews(NewsRequestTo
getNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(getNewsRequestTo);

            News demandedNews = await
_newsRepository.GetByIdAsync(newsFromDto.Id) ?? throw new
NewNotFoundException($"Demanded news {newsFromDto} was not found");

            NewsResponseTo demandedNewsResponseTo =
_mapper.Map<NewsResponseTo>(demandedNews);

            return demandedNewsResponseTo;
```

```
        }

        public async Task<NewsResponseTo> UpdateNews(NewsRequestTo
updateNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(updateNewsRequestTo);

            News? updateNews = await _newsRepository.UpdateAsync(newsFromDto)
?? throw new NewNotFoundException($"Update news {newsFromDto} was not
found");

            NewsResponseTo updateNewsResponseTo =
_mapper.Map<NewsResponseTo>(updateNews);

            return updateNewsResponseTo;
        }
    }

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;

public class NoteService : INoteService
    {
        private readonly IMapper _mapper;
        private readonly INoteRepository _noteRepository;

        public NoteService(IMapper mapper, INoteRepository repository)
        {
            _mapper = mapper;
            _noteRepository = repository;
        }

        public async Task<NoteResponseTo> CreateNote(NoteRequestTo
```

```
createNoteRequestTo)
    {
        Note noteFromDto = _mapper.Map<Note>(createNoteRequestTo);

        Note createdNote = await _noteRepository.AddAsync(noteFromDto);

        NoteResponseTo dtoFromCreatedNote =
            _mapper.Map<NoteResponseTo>(createdNote);

        return dtoFromCreatedNote;
    }

    public async Task DeleteNote(NoteRequestTo deleteNoteRequestTo)
    {
        Note noteFromDto = _mapper.Map<Note>(deleteNoteRequestTo);

        _ = await _noteRepository.DeleteAsync(noteFromDto)
            ?? throw new NoteNotFoundException(
                $"Delete note {noteFromDto} was not found");
    }

    public async Task<IEnumerable<NoteResponseTo>> GetAllNotes()
    {
        IEnumerable<Note> allNotes =
            await _noteRepository.GetAllAsync();

        var response = new List<NoteResponseTo>();

        foreach (Note note in allNotes)
        {
            response.Add(_mapper.Map<NoteResponseTo>(note));
        }

        return response;
    }

    public async Task<NoteResponseTo> GetNote(NoteRequestTo
getNoteRequestTo)
    {
        Note noteFromDto = _mapper.Map<Note>(getNoteRequestTo);
```

```
            Note demandedNote =
                await _noteRepository.GetByIdAsync(noteFromDto.Id)
                ?? throw new NoteNotFoundException(
                    $"Demanded note {noteFromDto} was not found");

            return _mapper.Map<NoteResponseTo>(demandedNote);
        }

        public async Task<NoteResponseTo> UpdateNote(NoteRequestTo
updateNoteRequestTo)
        {
            Note noteFromDto = _mapper.Map<Note>(updateNoteRequestTo);

            Note updatedNote =
                await _noteRepository.UpdateAsync(noteFromDto)
                ?? throw new NoteNotFoundException(
                    $"Update note {noteFromDto} was not found");

            return _mapper.Map<NoteResponseTo>(updatedNote);
        }
    }

using System.Linq.Expressions;
using rest1.application.interfaces;

namespace rest1.application.specifications;

public abstract class BaseSpecification<T> : ISpecification<T>
{
    public Expression<Func<T, bool>> Criteria { get; private set; }

    public List<Expression<Func<T, object>>> Includes { get; } = [];

    public Expression<Func<T, object>>? OrderBy { get; private set; }

    public Expression<Func<T, object>>? OrderByDescending { get; private set; }

    public int? Take { get; private set; }
```

```
    public int? Skip { get; private set; }

    public bool IsPagingEnabled { get; private set; }

    protected BaseSpecification(Expression<Func<T, bool>> criteria)
    {
        Criteria = criteria;
    }

    protected void AddInclude(Expression<Func<T, object>> includeExpression)
    {
        Includes.Add(includeExpression);
    }

    protected void ApplyPaging(int skip, int take)
    {
        Skip = skip;
        Take = take;
        IsPagingEnabled = true;
    }

    protected void ApplyOrderBy(Expression<Func<T, object>>
orderByExpression)
    {
        OrderBy = orderByExpression;
    }

    protected void ApplyOrderByDescending(Expression<Func<T, object>>
orderByDescendingExpression)
    {
        OrderByDescending = orderByDescendingExpression;
    }
}

namespace rest1.core.entities;

public class Creator(
    string login,
    string password,
    string firstname,
```

```
    string lastname) : Entity
{
    public string Login { get; set; } = login;

    public string Password { get; set; } = password;

    public string Firstname { get; set; } = firstname;

    public string Lastname { get; set; } = lastname;
}

namespace rest1.core.entities;

public abstract class Entity
{
    public long Id { get; set; }
}

namespace rest1.core.entities;

public class Mark(string name) : Entity
{
    public string Name { get; set; } = name;
    public long[] NewsIds { get; set; } = new long[0];
}

namespace rest1.core.entities;

public class News : Entity
{
    public long CreatorId { get; set; }

    public string Title { get; set; } = string.Empty;

    public string Content { get; set; } = string.Empty;

    public DateTime CreatedAt { get; set; }

    public DateTime Modified { get; set; }
```

```
    // lookup object
    public Creator? Creator { get; set; }
    public long[]? MarkIds { get; set; }
}

namespace rest1.core.entities;

public class Note : Entity
{
    public long NewsId { get; set; }

    public string Content { get; set; } = string.Empty;

    // lookup object
    public News? News { get; set; }


}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class CreatorRepository : Repository<Creator>, ICreatorRepository
{

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class MarkRepository: Repository<Mark>, IMarkRepository
{

}

using rest1.application.interfaces;
```

```
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class NewsRepository: Repository<News>, INewsRepository
{

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class NoteRepository: Repository<Note>, INoteRepository
{

}

using System.Collections.Concurrent;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class Repository<TEntity> : IRepository<TEntity> where TEntity : Entity
{
   protected readonly ConcurrentDictionary<long, TEntity> _entities = new();

   private long _nextId = 1;

   private readonly Lock _lock = new();

   public virtual Task<TEntity?> FindAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
   {
      cancellationToken.ThrowIfCancellationRequested();

      lock (_lock)
      {
```

```
            IQueryable<TEntity> query = ApplySpecification(specification);
            return Task.FromResult(query.FirstOrDefault());
        }
    }

    public virtual Task<IEnumerable<TEntity>>
FindAllAsync(ISpecification<TEntity> specification, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            IQueryable<TEntity> query = ApplySpecification(specification);
            return Task.FromResult(query.ToList().AsEnumerable());
        }
    }

    public virtual Task<int> CountAsync(ISpecification<TEntity> specification,
CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            IQueryable<TEntity> query = ApplySpecification(specification);
            return Task.FromResult(query.Count());
        }
    }

    public virtual Task<bool> AnyAsync(ISpecification<TEntity> specification,
CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            IQueryable<TEntity> query = ApplySpecification(specification);
            return Task.FromResult(query.Any());
        }
```

```
    }


    // CRUD
    public virtual Task<TEntity> AddAsync(TEntity entity, CancellationToken
cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            var id = entity.Id;
            if (id == 0)
            {
                id = _nextId;
            }

            if (_entities.TryAdd(_nextId, entity))
            {
                entity.Id = id;
                _nextId++;
                return Task.FromResult(entity);
            }

            throw new InvalidOperationException($"Entity with id {id} already
exists");
        }
    }

    public virtual Task<TEntity?> UpdateAsync(TEntity entity, CancellationToken
cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            var id = entity.Id;
```

```
            if (_entities.TryGetValue(id, out TEntity? updateEntity))
            {
                _entities[id] = entity;
                updateEntity = _entities[id];
            }

            return Task.FromResult(updateEntity);
        }
    }

    public virtual Task<TEntity?> DeleteAsync(TEntity entity, CancellationToken
cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            var id = entity.Id;
            _entities.TryRemove(id, out TEntity? deletedEntity);

            return Task.FromResult(deletedEntity);
        }
    }

    public virtual Task<bool> DeleteByIdAsync(long id, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            return Task.FromResult(_entities.TryRemove(id, out _));
        }
    }
    public virtual Task<TEntity?> GetByIdAsync(long id, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();
```

```
        lock (_lock)
        {
            return Task.FromResult(_entities.GetValueOrDefault(id));
        }
    }
    public virtual Task<IEnumerable<TEntity>> GetAllAsync(CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            return Task.FromResult(_entities.Values.AsEnumerable());
        }
    }
    protected virtual IQueryable<TEntity>
ApplySpecification(ISpecification<TEntity> spec)
    {
        IQueryable<TEntity> query = _entities.Values.AsQueryable();

        if (spec.Criteria != null)
        {
            query = query.Where(spec.Criteria);
        }
        if (spec.OrderBy != null)
        {
            query = query.OrderBy(spec.OrderBy);
        }
        else if (spec.OrderByDescending != null)
        {
            query = query.OrderByDescending(spec.OrderByDescending);
        }

        if (spec.IsPagingEnabled)
        {
            query = query.Skip(spec.Skip ?? 0)
                         .Take(spec.Take ?? 10);
        }
        return query;
    }
```

```
    public void Clear()
    {
        lock (_lock)
        {
            _entities.Clear();
        }
    }
}
```

```
using Microsoft.EntityFrameworkCore;
using rest1.core.entities;
using rest1.persistence.db.configurations;

namespace rest1.persistence.db;

public class RestServiceDbContext : DbContext
{
    public RestServiceDbContext(DbContextOptions<RestServiceDbContext>
options)
        : base(options) { }

    public DbSet<Creator> Creators { get; set; }

    public DbSet<Note> Notes { get; set; }

    public DbSet<News> News { get; set; }

    public DbSet<Mark> Marks { get; set; }

    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        modelBuilder.ApplyConfiguration(new CreatorConfiguration());
        modelBuilder.ApplyConfiguration(new MarkConfiguration());
        modelBuilder.ApplyConfiguration(new NoteConfiguration());
        modelBuilder.ApplyConfiguration(new NewsConfiguration());
        base.OnModelCreating(modelBuilder);
    }
}
```

```
using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;

public class CreatorConfiguration : IEntityTypeConfiguration<Creator>
{
   public void Configure(EntityTypeBuilder<Creator> builder)
   {
      builder.ToTable("tbl_creator", schema: "public");
      builder.Property(e => e.Id).HasColumnName("id");
      builder.Property(e => e.Login).HasColumnName("login");
      builder.Property(e => e.Password).HasColumnName("password");
      builder.Property(e => e.Firstname).HasColumnName("firstname");
      builder.Property(e => e.Lastname).HasColumnName("lastname");
   }
}

using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;

public class MarkConfiguration : IEntityTypeConfiguration<Mark>
{
   public void Configure(EntityTypeBuilder<Mark> builder)
   {
      builder.ToTable("tbl_mark", schema: "public");
      builder.Property(m => m.Id).HasColumnName("id");
      builder.Property(m => m.Name).HasColumnName("name");
   }
}

using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;
```

```
public class NewsConfiguration : IEntityTypeConfiguration<News>
{
   public void Configure(EntityTypeBuilder<News> builder)
   {
      builder.ToTable("tbl_news", schema: "public");
      builder.Property(n => n.Id).HasColumnName("id");
      builder.Property(n => n.CreatorId).HasColumnName("creator_id");
      builder.Property(n => n.Title).HasColumnName("title");
      builder.Property(n => n.Content).HasColumnName("content");
      builder.Property(n => n.CreatedAt).HasColumnName("created");
      builder.Property(n => n.Modified).HasColumnName("modified");
      builder.HasMany(n => n.Marks)
         .WithMany(m => m.News)
         .UsingEntity<Dictionary<string, object>>("tbl_newsMark",
            j => j.HasOne<Mark>()
               .WithMany()
               .HasForeignKey("MarkId")
               .HasConstraintName("FK_MarkNews_Mark"),
            j => j.HasOne<News>()
               .WithMany()
               .HasForeignKey("NewsId")
               .HasConstraintName("FK_MarkNews_News")
               .OnDelete(DeleteBehavior.Cascade),
            j => j.HasKey("MarkId", "NewsId")
         );
   }
}

using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;

public class NoteConfiguration : IEntityTypeConfiguration<Note>
{
   public void Configure(EntityTypeBuilder<Note> builder)
   {
      builder.ToTable("tbl_note", schema: "public");
```

```
        builder.Property(p => p.Id).HasColumnName("id");
        builder.Property(p => p.NewsId).HasColumnName("news_id");
        builder.Property(p => p.Content).HasColumnName("content");
    }
}

using Microsoft.EntityFrameworkCore;
using rest1.application.exceptions.db;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;

public class DbCreatorRepository : DbRepository<Creator>, ICreatorRepository
{
    public DbCreatorRepository(RestServiceDbContext dbContext) :
base(dbContext)
    {
    }

    public override async Task<Creator> AddAsync(Creator entity,
CancellationToken cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        var existingEntity = await _dbSet.Where(e => e.Id == entity.Id || e.Login ==
entity.Login).FirstOrDefaultAsync(cancellationToken);

        if (existingEntity != null)
        {
            throw new InvalidOperationException($"Entity with id {entity.Id} already
exists");
        }

        var entry = await _dbSet.AddAsync(entity, cancellationToken);
        try
        {
            await _dbContext.SaveChangesAsync(cancellationToken);
        }
```

```
        catch (DbUpdateException ex) when (IsForeignKeyViolation(ex))
        {
            throw new ForeignKeyViolationException("Foreign key doesn't exist");
        }

        return entry.Entity;
    }
}

using Microsoft.EntityFrameworkCore;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;

public class DbMarkRepository : DbRepository<Mark>, IMarkRepository
{
    public DbMarkRepository(RestServiceDbContext dbContext) : base(dbContext)
    {
    }

    public async Task DeleteMarksWithoutNews()
    {
        var emptyMarkers = await _dbSet.Where(m => m.News.Count ==
0).ToListAsync();
        _dbSet.RemoveRange(emptyMarkers);
        await _dbContext.SaveChangesAsync();
    }
}

using Microsoft.EntityFrameworkCore;
using rest1.application.exceptions.db;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;

 public class DbNewsRepository : DbRepository<News>, INewsRepository
{
    public DbNewsRepository(RestServiceDbContext dbContext) : base(dbContext)
```

```
    {

    }

    public override async Task<News> AddAsync(News entity, CancellationToken
cancellationToken = default)
    {

        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        var existingEntity = await _dbSet.Where(n => n.Id == entity.Id || n.Title ==
entity.Title).FirstOrDefaultAsync(cancellationToken);

        if (existingEntity != null)
        {
            throw new InvalidOperationException($"Entity with id {entity.Id} already
exists");
        }

        var entry = await _dbSet.AddAsync(entity, cancellationToken);
        try
        {
            await _dbContext.SaveChangesAsync(cancellationToken);
        }
        catch (DbUpdateException ex) when (IsForeignKeyViolation(ex))
        {
            throw new ForeignKeyViolationException("Foreign key doesn't exist");
        }

        return entry.Entity;
    }

    public async override Task<News?> DeleteAsync(News entity,
CancellationToken cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        var existingEntity = await _dbSet.Where(n => n.Id == entity.Id)
```

```
            .Include(n => n.Marks)
            .FirstOrDefaultAsync();

        if (existingEntity == null)
            return null;

        _dbSet.Remove(existingEntity);

        await _dbContext.SaveChangesAsync(cancellationToken);

        return existingEntity;
    }

    public async Task<News> GetByIdWithMarksAsync(int id)
    {
        return await _dbSet
            .Include(n => n.Marks)
            .FirstOrDefaultAsync(n => n.Id == id);
    }

    public async Task RemoveAsync(News news)
    {
        _dbSet.Remove(news);
        await _dbContext.SaveChangesAsync();
    }
}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;

public class DbNoteRepository : DbRepository<Note>, INoteRepository
{
    public DbNoteRepository(RestServiceDbContext dbContext) : base(dbContext)
    {
    }
}

using Microsoft.EntityFrameworkCore;
```

```
using Npgsql;
using rest1.application.exceptions.db;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;


 public class DbRepository<TEntity> : IRepository<TEntity>
       where TEntity : Entity
   {
       protected readonly DbSet<TEntity> _dbSet;
       protected readonly DbContext _dbContext;

       public DbRepository(RestServiceDbContext dbContext)
       {
           _dbContext = dbContext ?? throw new
ArgumentNullException(nameof(dbContext));
           _dbSet = dbContext.Set<TEntity>();
       }

       public virtual async Task<TEntity?> FindAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
       {
           cancellationToken.ThrowIfCancellationRequested();

           IQueryable<TEntity> query = ApplySpecification(specification);
           return await query.FirstOrDefaultAsync(cancellationToken);
       }

       public virtual async Task<IEnumerable<TEntity>>
FindAllAsync(ISpecification<TEntity> specification, CancellationToken
cancellationToken = default)
       {
           cancellationToken.ThrowIfCancellationRequested();

           IQueryable<TEntity> query = ApplySpecification(specification);
           return await query.ToListAsync(cancellationToken);
       }
```

```
    public virtual async Task<int> CountAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        IQueryable<TEntity> query = ApplySpecification(specification);
        return await query.CountAsync(cancellationToken);
    }

    public virtual async Task<bool> AnyAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        IQueryable<TEntity> query = ApplySpecification(specification);
        return await query.AnyAsync(cancellationToken);
    }

    // CRUD
    public virtual async Task<TEntity> AddAsync(TEntity entity,
CancellationToken cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        var existingEntity = await _dbSet.FindAsync([entity.Id],
cancellationToken: cancellationToken);

        if (existingEntity != null)
        {
            throw new InvalidOperationException($"Entity with id {entity.Id}
already exists");
        }

        var entry = await _dbSet.AddAsync(entity, cancellationToken);
        try
        {
            await _dbContext.SaveChangesAsync(cancellationToken);
        }
        catch (DbUpdateException ex) when (IsForeignKeyViolation(ex))
```

```
            {
                throw new ForeignKeyViolationException("Foreign key doesn't exist");
            }

            return entry.Entity;
        }

        protected static bool IsForeignKeyViolation(DbUpdateException ex)
        {
            return ex.InnerException switch
            {
                PostgresException pgEx => pgEx.SqlState == "23503",
                _ => false
            };
        }

        public virtual async Task<TEntity?> UpdateAsync(TEntity entity,
CancellationToken cancellationToken = default)
        {
            ArgumentNullException.ThrowIfNull(entity);
            cancellationToken.ThrowIfCancellationRequested();

            var existingEntity = await _dbSet.FindAsync([entity.Id,
cancellationToken], cancellationToken: cancellationToken);

            if (existingEntity == null)
                return null;

            _dbContext.Entry(existingEntity).CurrentValues.SetValues(entity);
            await _dbContext.SaveChangesAsync(cancellationToken);

            return existingEntity;
        }

        public virtual async Task<TEntity?> DeleteAsync(TEntity entity,
CancellationToken cancellationToken = default)
        {
            ArgumentNullException.ThrowIfNull(entity);
            cancellationToken.ThrowIfCancellationRequested();
```

```
            var existingEntity = await _dbSet.FindAsync(entity.Id, cancellationToken);

            if (existingEntity == null)
                return null;

            _dbSet.Remove(existingEntity);
            await _dbContext.SaveChangesAsync(cancellationToken);

            return existingEntity;
        }

        public virtual async Task<bool> DeleteByIdAsync(long id,
CancellationToken cancellationToken = default)
        {
            cancellationToken.ThrowIfCancellationRequested();

            var entity = await _dbSet.FindAsync(id , cancellationToken);

            if (entity == null)
                return false;

            _dbSet.Remove(entity);
            await _dbContext.SaveChangesAsync(cancellationToken);

            return true;
        }

        public virtual async Task<TEntity?> GetByIdAsync(long id,
CancellationToken cancellationToken = default)
        {
            cancellationToken.ThrowIfCancellationRequested();

            return await _dbSet.FindAsync(id, cancellationToken);
        }

        public virtual async Task<IEnumerable<TEntity>>
GetAllAsync(CancellationToken cancellationToken = default)
        {
            cancellationToken.ThrowIfCancellationRequested();
```

```
            return await _dbSet.ToListAsync(cancellationToken);
        }

        protected virtual IQueryable<TEntity>
ApplySpecification(ISpecification<TEntity> spec)
        {
            IQueryable<TEntity> query = _dbSet.AsQueryable();

            if (spec.Criteria != null)
            {
                query = query.Where(spec.Criteria);
            }

            // Применяем инклюды (включаемые навигационные свойства)
            query = spec.Includes
                .Aggregate(query, (current, include) => current.Include(include));

            if (spec.OrderBy != null)
            {
                query = query.OrderBy(spec.OrderBy);
            }
            else if (spec.OrderByDescending != null)
            {
                query = query.OrderByDescending(spec.OrderByDescending);
            }

            if (spec.IsPagingEnabled)
            {
                query = query.Skip(spec.Skip ?? 0)
                             .Take(spec.Take ?? 10);
            }

            return query;
        }
    }

using Cassandra;
using Cassandra.Mapping;

namespace DiscussionModule.persistence;
```

```
public class CassandraContext
{
    private readonly ICluster _cluster;
    private Cassandra.ISession _session;
    private IMapper _mapper;

    public CassandraContext(IConfiguration configuration)
    {
        var connectionString =
configuration.GetConnectionString("CassandraConnection");
        var options = new CassandraOptions(connectionString);

        _cluster = Cluster.Builder()
            .AddContactPoint(options.Host)
            .WithPort(options.Port)
            .Build();

        // Сначала подключаемся без указания keyspace
        _session = _cluster.Connect();
    }

    public IMapper Mapper => _mapper;
    public Cassandra.ISession Session => _session;

    public void CreateKeyspaceIfNotExists(string keyspaceName)
    {
        var createKeyspaceQuery = $@"
            CREATE KEYSPACE IF NOT EXISTS {keyspaceName}
            WITH replication = {{
                'class': 'SimpleStrategy',
                'replication_factor': 1
            }}";

        _session.Execute(createKeyspaceQuery);

        // Переключаемся на созданный keyspace
        _session.ChangeKeyspace(keyspaceName);

        // Создаем mapper после переключения keyspace
```

```
        _mapper = new Mapper(_session);
    }

    public void CreateTableIfNotExists()
    {
        var createTableQuery = @"
            CREATE TABLE IF NOT EXISTS tbl_notes (
                id bigint PRIMARY KEY,
                news_id bigint,
                content text,
                country text
            )";

        _session.Execute(createTableQuery);
    }

    public void CreateCounterTableIfNotExists()
    {
        _session.Execute("DROP TABLE IF EXISTS tbl_counters");
        var createCounterTableQuery = @"
        CREATE TABLE IF NOT EXISTS tbl_counters (
            counter_name text PRIMARY KEY,
            counter_value counter
        )";

        _session.Execute(createCounterTableQuery);

        var initCounterQuery = @"
        UPDATE tbl_counters
        SET counter_value = counter_value + 0
        WHERE counter_name = 'note_id'";

        _session.Execute(initCounterQuery);
    }
}

public class CassandraOptions
{
    public string Host { get; }
    public int Port { get; }
```

```
    public string Keyspace { get; }

    public CassandraOptions(string connectionString)
    {
        // Парсинг connection string формата:
jdbc:cassandra://localhost:9042/distcompcasssandra
        var uri = new Uri(connectionString.Replace("jdbc:", ""));
        Host = uri.Host;
        Port = uri.Port;
        Keyspace = uri.AbsolutePath.TrimStart('/');
    }
}

using Cassandra;
using Cassandra.Mapping;
using DiscussionModule.interfaces;
using DiscussionModule.models;

namespace DiscussionModule.persistence.repositories;

public class NoteRepository : INoteRepository
{
    private readonly IMapper _mapper;
    private readonly Cassandra.ISession _session;

    public NoteRepository(CassandraContext context)
    {
        _mapper = context.Mapper;
        _session = context.Session;
    }

    public async Task<Note> AddAsync(Note note)
    {
        note.Id = await GetNextId();

        await _mapper.InsertAsync(note);
        return note;
    }

    public async Task<Note?> GetByIdAsync(long id)
```

```
{
    var note = await _mapper.SingleOrDefaultAsync<Note>(
        "WHERE id = ?", id);
    return note;
}

public async Task<List<Note>> GetAllAsync()
{
    var notes = await _mapper.FetchAsync<Note>();
    return notes.ToList();
}

public async Task<Note?> UpdateAsync(Note note)
{
    var existing = await GetByIdAsync(note.Id);
    if (existing == null)
        return null;

    await _mapper.UpdateAsync(note);
    return note;
}

public async Task DeleteAsync(Note note)
{
    await _mapper.DeleteAsync(note);
}

private async Task<long> GetNextId()
{
    try
    {
        var updateQuery = @"
            UPDATE tbl_counters
            SET counter_value = counter_value + 1
            WHERE counter_name = 'note_id'";

        await _session.ExecuteAsync(new SimpleStatement(updateQuery));

        var selectQuery = @"
            SELECT counter_value FROM tbl_counters
```

```
            WHERE counter_name = 'note_id'";

        var row = (await _session.ExecuteAsync(
            new SimpleStatement(selectQuery))).FirstOrDefault();

        if (row != null)
        {
            return row.GetValue<long>("counter_value");
        }

        var initQuery = @"
            UPDATE tbl_counters
            SET counter_value = counter_value + 0
            WHERE counter_name = 'note_id'";

        await _session.ExecuteAsync(new SimpleStatement(initQuery));

        var initRow = (await _session.ExecuteAsync(
            new SimpleStatement(selectQuery))).FirstOrDefault();

        return initRow?.GetValue<long>("counter_value") ?? 0;
      }
      catch (Exception ex)
      {
        throw new Exception("Ошибка при генерации ID для заметки", ex);
      }
    }
}

using DiscussionModule.DTOs.requests;
using DiscussionModule.DTOs.responses;
using DiscussionModule.interfaces;
using Microsoft.AspNetCore.Mvc;

namespace DiscussionModule.controllers;

[ApiController]
[Route("api/v1.0/[controller]")]
public class NotesController : ControllerBase
{
```

```
    private readonly INoteService _service;

    public NotesController(INoteService service)
    {
        _service = service;
    }

    [HttpPost]
    public async Task<ActionResult<NoteResponseTo>> CreateNote([FromBody]
NoteRequestTo note)
    {
        var created = await _service.CreateNote(note);
        return CreatedAtAction(nameof(GetNoteById), new { id = created.Id },
created);
    }

    [HttpGet]
    public async Task<ActionResult<IEnumerable<NoteResponseTo>>>
GetAllNotes()
    {
        try
        {
            var notes = await _service.GetAllNotes();
            return Ok(notes);
        }
        catch(Exception)
        {
            return Ok();
        }
    }

    [HttpGet("{id:long}")]
    public async Task<ActionResult<NoteResponseTo>> GetNoteById(long id)
    {
        var dto = new NoteRequestTo() { Id = id };
        var post = await _service.GetNote(dto);
        if (post == null)
            return NotFound();

        return Ok(post);
```

```
    }

    [HttpPut("{id:long}")]
    public async Task<ActionResult<NoteResponseTo>> UpdateNote(long id,
[FromBody] NoteRequestTo note)
    {
        note.Id = id;
        var updated = await _service.UpdateNote(note);
        if (updated == null)
            return NotFound();

        return Ok(updated);
    }

    [HttpDelete("{id:long}")]
    public async Task<IActionResult> DeleteNote(long id)
    {
        var dto = new NoteRequestTo() { Id = id };
        await _service.DeleteNote(dto);

        return NoContent();
    }
}

using Cassandra.Mapping.Attributes;

namespace DiscussionModule.models;

[Table("tbl_notes")]
public class Note
{
    [PartitionKey]
    [Column("id")]
    public long Id { get; set; }

    [Column("news_id")]
    public long NewsId { get; set; }

    [Column("content")]
    public string Content { get; set; } = string.Empty;
```

```
    [Column("country")]
    public string Country { get; set; } = string.Empty;
}

using Confluent.Kafka;
using System.Text.Json;
using DiscussionModule.interfaces;
using DiscussionModule.DTOs.requests;
using AutoMapper;

namespace DiscussionModule.kafka;

public class KafkaConsumer : BackgroundService
{
    private readonly IConsumer<string, string> _consumer;
    private readonly IServiceProvider _serviceProvider;
    private readonly string _inTopic;
    private readonly KafkaProducer _producer;
    private readonly List<string> _stopWords = new() { "spam", "bad", "offensive",
"xxx" };

    public KafkaConsumer(IServiceProvider serviceProvider, string
bootstrapServers, string inTopic, string outTopic)
    {
        var config = new ConsumerConfig
        {
            BootstrapServers = bootstrapServers,
            GroupId = "discussion-consumer-group",
            AutoOffsetReset = AutoOffsetReset.Earliest,
            EnableAutoCommit = true
        };

        _consumer = new ConsumerBuilder<string, string>(config).Build();
        _serviceProvider = serviceProvider;
        _inTopic = inTopic;
        _producer = new KafkaProducer(bootstrapServers, outTopic);
    }

    protected override async Task ExecuteAsync(CancellationToken
```

```
stoppingToken)
    {
        _consumer.Subscribe(_inTopic);
        Console.WriteLine($"[Kafka] Consumer started, listening to topic:
{_inTopic}");

        while (!stoppingToken.IsCancellationRequested)
        {
            try
            {
                var consumeResult = _consumer.Consume(stoppingToken);

                if (consumeResult != null)
                {
                    await ProcessMessageAsync(consumeResult.Message.Key,
consumeResult.Message.Value);
                }
            }
            catch (OperationCanceledException)
            {
                break;
            }
            catch (Exception ex)
            {
                Console.WriteLine($"[Kafka] Error: {ex.Message}");
            }
        }

        _consumer.Close();
    }

    private async Task ProcessMessageAsync(string key, string jsonMessage)
    {
        using var scope = _serviceProvider.CreateScope();
        var noteService =
scope.ServiceProvider.GetRequiredService<INoteService>();
        var mapper = scope.ServiceProvider.GetRequiredService<IMapper>();
        var operation = key;
        var noteRequest = JsonSerializer.Deserialize<NoteRequestTo>(jsonMessage);
```

```
try
{
    object response = null;
    string status = "SUCCESS";

    switch (operation.ToUpper())
    {
        case "CREATE":
            if (await ModerateNoteAsync(noteRequest))
            {
                var createdNote = await noteService.CreateNote(noteRequest);
                response = createdNote;
                status = "APPROVED";
            }
            else
            {
                status = "DECLINED";
                response = new { Message = "Note declined by moderation" };
            }
            break;

        case "GET":
            response = await noteService.GetNote(noteRequest);
            break;

        case "UPDATE":
            response = await noteService.UpdateNote(noteRequest);
            break;

        case "DELETE":
            await noteService.DeleteNote(noteRequest);
            response = new { Message = "Deleted successfully" };
            break;
    }

    var responseMessage = new
    {
        Operation = operation,
        Status = status,
        Data = response,
```

```
            RequestId = noteRequest?.Id ?? 0
        };

        await _producer.SendMessageAsync(noteRequest?.Id.ToString() ?? "0",
responseMessage);
    }
    catch (Exception ex)
    {
        var errorResponse = new
        {
            Operation = key,
            Status = "ERROR",
            Error = ex.Message,
            RequestId = noteRequest?.Id ?? 0
        };

        await _producer.SendMessageAsync(noteRequest?.Id.ToString() ?? "0",
errorResponse);
        Console.WriteLine($"[Kafka] Error processing: {ex.Message}");
    }
}

private Task<bool> ModerateNoteAsync(NoteRequestTo note)
{

    // Проверка стоп-слов
    foreach (var stopWord in _stopWords)
    {
        if (note.Content.ToLower().Contains(stopWord))
        {
            return Task.FromResult(false);
        }
    }

    return Task.FromResult(true);
}

public override void Dispose()
{
    _consumer?.Dispose();
```

```
        _producer?.Dispose();
        base.Dispose();
    }
}

using Confluent.Kafka;
using System.Text.Json;
using Confluent.Kafka.Admin;

namespace DiscussionModule.kafka;

public class KafkaProducer : IDisposable
{
    private readonly IProducer<string, string> _producer;
    private readonly string _outTopic;
    private readonly string _bootstrapServers;

    public KafkaProducer(string bootstrapServers, string outTopic)
    {
        _bootstrapServers = bootstrapServers;
        var config = new ProducerConfig
        {
            BootstrapServers = bootstrapServers,
            ClientId = "discussion-module-producer",
            Acks = Acks.All,
            EnableIdempotence = true
        };

        _producer = new ProducerBuilder<string, string>(config).Build();
        _outTopic = outTopic;
    }

    public async Task SendMessageAsync(string key, object message)
    {
        try
        {
            var jsonMessage = JsonSerializer.Serialize(message);
            var result = await _producer.ProduceAsync(_outTopic, new
Message<string, string>
            {
```

```
            Key = key,
            Value = jsonMessage
        });

    }
    catch (ProduceException<string, string> ex)
    {
        Console.WriteLine($"[Kafka] Error: {ex.Error.Reason}");
        throw;
    }
}

public async Task EnsureTopicsExistAsync(params string[] topics)
{
    try
    {
        var config = new AdminClientConfig
        {
            BootstrapServers = _bootstrapServers
        };

        using var adminClient = new AdminClientBuilder(config).Build();

        var topicSpecifications = topics.Select(topic => new TopicSpecification
        {
            Name = topic,
            NumPartitions = 3,
            ReplicationFactor = 1
        }).ToList();

        try
        {
            await adminClient.CreateTopicsAsync(topicSpecifications);
            Console.WriteLine($"[Kafka] Topics created: {string.Join(", ",
topics)}");
        }
        catch (CreateTopicsException ex)
        {
            if (!ex.Results.Any(r => r.Error.Code !=
ErrorCode.TopicAlreadyExists))
```

```
            {
                Console.WriteLine("[Kafka] Topics already exist");
            }
            else
            {
                Console.WriteLine($"[Kafka] Error: {ex.Message}");
            }
        }
    }
    catch (Exception ex)
    {
        Console.WriteLine($"[Kafka] Warning: {ex.Message}");
    }
}

public void Dispose()
{
    _producer?.Dispose();
}
}

using DiscussionModule.interfaces;
using DiscussionModule.kafka;
using DiscussionModule.mappers;
using DiscussionModule.persistence;
using DiscussionModule.persistence.repositories;
using DiscussionModule.services;

var builder = WebApplication.CreateBuilder(args);

// Add services to the container
builder.Services.AddControllers();

// Register CassandraContext as singleton
builder.Services.AddSingleton<CassandraContext>(serviceProvider =>
{
    var configuration = serviceProvider.GetRequiredService<IConfiguration>();
    var context = new CassandraContext(configuration);

    context.CreateKeyspaceIfNotExists("distcompcasssandra");
```

```
    context.CreateTableIfNotExists();
    context.CreateCounterTableIfNotExists();

    return context;
});

builder.Services.AddScoped<INoteRepository, NoteRepository>();
builder.Services.AddScoped<INoteService, NoteService>();

// AutoMapper
builder.Services.AddAutoMapper(config =>
{
    config.AddProfile<NoteProfile>();
});

// === KAFKA CONFIGURATION ===
var kafkaEnabled = builder.Configuration.GetValue<bool>("Kafka:Enabled");

if (kafkaEnabled)
{
    var kafkaBootstrapServers =
builder.Configuration.GetValue<string>("Kafka:BootstrapServers") ??
"localhost:9092";
    var kafkaInTopic = builder.Configuration.GetValue<string>("Kafka:InTopic")
?? "InTopic";
    var kafkaOutTopic =
builder.Configuration.GetValue<string>("Kafka:OutTopic") ?? "OutTopic";

    builder.Services.AddSingleton<KafkaProducer>(provider =>
        new KafkaProducer(kafkaBootstrapServers, kafkaOutTopic));

    builder.Services.AddSingleton<KafkaConsumer>(provider =>
        new KafkaConsumer(
            provider,
            kafkaBootstrapServers,
            kafkaInTopic,
            kafkaOutTopic
        ));

    builder.Services.AddHostedService(provider =>
```

```
        provider.GetRequiredService<KafkaConsumer>());
}

var app = builder.Build();

if (kafkaEnabled)
{
    using var scope = app.Services.CreateScope();
    var producer = scope.ServiceProvider.GetRequiredService<KafkaProducer>();

    await producer.EnsureTopicsExistAsync(
        builder.Configuration["Kafka:InTopic"],
        builder.Configuration["Kafka:OutTopic"]
    );
}
app.MapControllers();
app.Run();


namespace RedisService.interfaces;

public interface IRedisCacheService
{
    Task<T?> GetAsync<T>(string key);
    Task SetAsync<T>(string key, T value, TimeSpan? expiry = null);
    Task RemoveAsync(string key);
    Task<bool> ExistsAsync(string key);
    Task RemoveByPatternAsync(string pattern);
    Task<T?> GetOrSetAsync<T>(string key, Func<Task<T>> factory, TimeSpan?
expiry = null);
}


using System.Text.Json;
using Microsoft.Extensions.Configuration;
using Microsoft.Extensions.Logging;
using RedisService.interfaces;
using StackExchange.Redis;

namespace RedisService.services;
```

```
public class RedisCacheService : IRedisCacheService
{
    private readonly IConnectionMultiplexer _redis;
    private readonly IDatabase _database;
    private readonly ILogger<RedisCacheService> _logger;
    private readonly TimeSpan _defaultExpiry;

    public RedisCacheService(IConfiguration configuration,
ILogger<RedisCacheService> logger)
    {
        var connectionString =
configuration.GetConnectionString("RedisConnection")
            ?? "localhost:6379";

        _redis = ConnectionMultiplexer.Connect(connectionString);
        _database = _redis.GetDatabase();
        _logger = logger;

        int expiryMinutes = 5;
        _defaultExpiry = TimeSpan.FromMinutes(expiryMinutes);
    }

    public async Task<T?> GetAsync<T>(string key)
    {
        try
        {
            var value = await _database.StringGetAsync(key);

            if (value.IsNull)
                return default;

            return JsonSerializer.Deserialize<T>(value!);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting value from Redis for key: {Key}",
key);
            return default;
        }
```

```
    }

    public async Task SetAsync<T>(string key, T value, TimeSpan? expiry = null)
    {
        try
        {
            var serializedValue = JsonSerializer.Serialize(value);
            await _database.StringSetAsync(key, serializedValue, expiry ??
_defaultExpiry);

            _logger.LogDebug("Value cached with key: {Key}", key);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error setting value in Redis for key: {Key}", key);
        }
    }

    public async Task RemoveAsync(string key)
    {
        try
        {
            await _database.KeyDeleteAsync(key);
            _logger.LogDebug("Key removed from cache: {Key}", key);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error removing key from Redis: {Key}", key);
        }
    }

    public async Task<bool> ExistsAsync(string key)
    {
        try
        {
            return await _database.KeyExistsAsync(key);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error checking key existence in Redis: {Key}",
```

```
key);
            return false;
        }
    }

    public async Task RemoveByPatternAsync(string pattern)
    {
        try
        {
            var server = _redis.GetServer(_redis.GetEndPoints().First());
            var keys = server.Keys(pattern: pattern).ToArray();

            if (keys.Any())
            {
                await _database.KeyDeleteAsync(keys);
                _logger.LogDebug("Removed {Count} keys matching pattern:
{Pattern}", keys.Length, pattern);
            }
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error removing keys by pattern from Redis:
{Pattern}", pattern);
        }
    }

    public async Task<T?> GetOrSetAsync<T>(string key, Func<Task<T>>
factory, TimeSpan? expiry = null)
    {
        var cached = await GetAsync<T>(key);

        if (cached != null)
        {
            _logger.LogDebug("Cache hit for key: {Key}", key);
            return cached;
        }

        _logger.LogDebug("Cache miss for key: {Key}", key);

        var result = await factory();
```

```
        if (result != null)
        {
            await SetAsync(key, result, expiry);
        }

        return result;
    }
}
```